PROJECT - 1

Decoding Parameters & Sampling Analysis

Local LLM Benchmarking with Ollama



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Course	Special topics in AI
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Primary Model	qwen-seo:4b
Course Instructor	Sir. Shaban Ghias Satti

1. Objective

The purpose of this project is to study how decoding and sampling parameters affect the output of a local Large Language Model (LLM). The same test prompt will be sent to the model several times while changing Temperature, Top-K and Top-P. The generated responses will then be compared manually for determinism, creativity, wording, structure and relevance.

The primary model used in this project is qwen-seo:4b, which is already downloaded and available locally through Ollama.

2. Project Setup

Step 1 - Create a Project Folder

```
mkdir SpecialTopicsinAI
cd SpecialTopicsinAI
```

A separate project folder keeps the Python file, virtual environment and configuration file organized.

Step 2 - Create a Virtual Environment

```
python -m venv venv
```

A virtual environment keeps the libraries used by this project separate from other Python projects on the computer.

Step 3 - Activate the Virtual Environment

```
venv\Scripts\activate
```

On Windows, this command activates the virtual environment. After activation, the command prompt normally shows (venv) before the folder path.

Step 4 - Create a .env File

Create a file named .env in the project folder and paste the local Qwen/Ollama URL and model name:

```
QWEN_URL=http://localhost:11434/api/generate
MODEL_NAME=qwen-seo:4b
```

The .env file stores configuration values outside the Python code. This makes the program cleaner and allows the URL or model name to be changed without editing the main Python file.

Step 5 - Confirm the Model

```
ollama list
```

Because qwen-seo:4b is already downloaded, this step only confirms that it is available in Ollama.

If the model is missing on another computer, the download command is simply:

```
ollama pull qwen-seo:4b
```

Step 6 – Create Requirements.txt

Add library names we want to install:

requests
pandas
python-dotenv

Then run following command to install all added libraries at once.

```
pip install -r requirements.txt
```

3. Libraries Used and Why

Library	Purpose	Why it is used
requests	Communicates with Ollama	Sends the model name, prompt and sampling parameters to the local Ollama API.
pandas	Displays experiment results	Creates a simple comparison table inside Python. No CSV file is generated.
python-dotenv	Reads the .env file	Loads QWEN_URL and MODEL_NAME without hard-coding them in the Python file.
time	Measures response time	Records how long each model response takes. It is built into Python.
os	Reads environment variables	Gets the values loaded from the .env file. It is built into Python.

4. Sampling Parameters

4.1 Temperature

Temperature controls randomness in token selection. A low value makes the model more predictable and focused. A higher value allows more variation and creativity.

Temperature	Expected Effect	Simple Meaning
0.2	Very focused and consistent	Low randomness
0.7	Balanced response	Moderate randomness
1.1	More diverse and creative	Higher randomness

4.2 Top-K

Top-K limits the next-token choices to the K most probable tokens. A small Top-K gives the model fewer choices, while a larger Top-K allows more possible tokens to be considered.

Top-K	Expected Effect	Simple Meaning
10-20	More restricted and stable	Few token choices
40	Balanced	Moderate token choices
80-100	More varied	More token choices

4.3 Top-P

Top-P, also called nucleus sampling, selects tokens from the smallest group whose combined probability reaches the selected value. A lower Top-P makes output more conservative, while a higher Top-P allows a wider range of likely tokens.

Top-P	Expected Effect	Simple Meaning
0.50-0.70	More controlled	Narrow probability range
0.90	Balanced	Common practical setting
0.98-0.99	More diverse	Wide probability range

4.4 Combined Effect

Temperature, Top-K and Top-P work together. Lower values usually make the response more deterministic. Higher values generally increase diversity, but too much randomness can reduce consistency or introduce irrelevant details.

5. Step-by-Step Python Code

Step 1 - Import Libraries

```
import os
import time
import requests
import pandas as pd
from dotenv import load_dotenv
```

These imports allow the program to load configuration values, call Ollama, measure time and display results.

Step 2 - Load Values from .env

```
load_dotenv()

QWEN_URL = os.getenv("QWEN_URL")
MODEL_NAME = os.getenv("MODEL_NAME")
```

load_dotenv() reads the .env file. os.getenv() then retrieves the Ollama URL and model name.

Step 3 - Create One Fixed Test Prompt

```
TEST_PROMPT = (
    "Explain why keyword research is important for SEO. "
    "Give one short paragraph followed by three practical benefits."
)
```

The same prompt must be used in every experiment so that output differences can be linked to the sampling settings.

Step 4 - Create a Function to Generate a Response

```
def generate_response(prompt, temperature, top_k, top_p):
    payload = {
        "model": MODEL_NAME,
        "prompt": prompt,
        "stream": False,
        "options": {
            "temperature": temperature,
            "top_k": top_k,
            "top_p": top_p
        }
    }

    start_time = time.time()

    response = requests.post(
        QWEN_URL,
        json=payload,
        timeout=300
    )

    end_time = time.time()

    response.raise_for_status()
    result = response.json()

    return result["response"], round(end_time - start_time, 2)
```

This function sends the prompt and decoding settings to qwen-seo:4b through Ollama. It returns the response and the generation time.

Step 5 - Define Three Main Sampling Profiles

```
experiments = [
    {
        "setting": "Conservative",
        "temperature": 0.2,
        "top_k": 20,
        "top_p": 0.70
    },
    {
        "setting": "Balanced",
        "temperature": 0.7,
        "top_k": 40,
        "top_p": 0.90
    },
    {
        "setting": "Creative",
        "temperature": 1.1,
        "top_k": 80,
        "top_p": 0.98
    }
]
```

These profiles give a simple low, medium and high-randomness comparison.

Step 6 - Run the Experiment

```
results = []

for experiment in experiments:
    output, response_time = generate_response(
        TEST_PROMPT,
        experiment["temperature"],
        experiment["top_k"],
        experiment["top_p"]
    )

    results.append({
        "Setting": experiment["setting"],
        "Temperature": experiment["temperature"],
        "Top-K": experiment["top_k"],
        "Top-P": experiment["top_p"],
        "Time (seconds)": response_time,
        "Response": output
    })

    print("\n" + "=" * 70)
    print("Setting:", experiment["setting"])
    print("Temperature:", experiment["temperature"])
    print("Top-K:", experiment["top_k"])
    print("Top-P:", experiment["top_p"])
    print("Time:", response_time, "seconds")
    print("\nResponse:")
    print(output)
```

Each setting uses the same prompt. The full response is printed directly so screenshots can be taken and the outputs can be compared manually.

Step 7 - Display a Simple On-Screen Table

```
df = pd.DataFrame(results)

print("\nExperiment Results:")
print(df[[
    "Setting",
    "Temperature",
    "Top-K",
    "Top-P",
    "Time (seconds)"
]])
```

This displays the experiment settings and response times inside Python. No CSV file is created.

8. Controlled Experiment

The main experiment changes several parameters together. A controlled experiment changes only one parameter at a time while keeping the other two fixed. This makes the individual effect easier to observe.

8.1 Temperature-Only Test

```
temperature_values = [0.2, 0.7, 1.1]

for value in temperature_values:
    output, response_time = generate_response(
        TEST_PROMPT,
        temperature=value,
        top_k=40,
        top_p=0.90
    )

    print("\nTemperature:", value)
    print("Time:", response_time, "seconds")
    print(output)
```

Expected effect: increasing Temperature should increase variation, while lower Temperature should be more stable and predictable.

Qwen Output :

==========
CONTROLLED EXPERIMENT: TEMPERATURE
Fixed Top-K = 40
Fixed Top-P = 0.90
==========

==========
Temperature: 0.2
Top-K: 40
Top-P: 0.90
Time: 74.04 seconds

Response:
<think>
Okay, the user wants an explanation of why keyword research is important for SEO. They need a short paragraph followed by three practical benefits. Let me start by recalling the main points about keyword research. It's about understanding what users are searching for, which helps in creating content that meets their needs. That's crucial for SEO because search engines use keywords to index pages.

Now, the first paragraph should be concise. I need to mention how keyword research helps align content with user intent and search engine algorithms. Maybe mention that it's the foundation of SEO strategies. Then, the three benefits. Let me think: first, it improves content relevance, so search engines can rank the site higher. Second, it helps in targeting specific audiences, which can lead to better engagement. Third, it optimizes for long-tail keywords, which have lower competition and higher intent. Wait, is that accurate? Yes, long-tail keywords are more specific and have

higher conversion rates. Also, it helps in identifying gaps in the content, so you can create more targeted content. Maybe that's the third benefit. Let me check if those are the top three. Alternatively, maybe the third is about improving user experience by aligning content with what users need. Hmm. Need to make sure the benefits are practical and directly tied to SEO outcomes. Let me structure them as: 1. Enhances content relevance and search engine rankings. 2. Targets specific audiences with tailored content. 3. Optimizes for long-tail keywords, increasing conversion potential. That sounds good. Need to make sure each benefit is clear and concise. Alright, that should cover it.
</think>

Keyword research is essential for SEO because it helps identify the specific terms and phrases users search for, aligning content with their intent and improving visibility in search engine results. It forms the foundation of any SEO strategy by ensuring content is optimized for relevant queries, enhancing both user experience and search engine rankings.

Practical Benefits:
1. **Improves Content Relevance:** Identifies high-intent keywords that align with user needs, making content more likely to be ranked for targeted searches.
2. **Targets Specific Audiences:** Reveals niche keywords that help tailor content to specific demographics or interests, increasing engagement and conversion.
3. **Optimizes for Long-Tail Keywords:** Focuses on less competitive, high-intent keywords that drive targeted traffic and improve conversion rates.

==========
Temperature: 0.7
Top-K: 40
Top-P: 0.90
Time: 80.05 seconds

Response:
<think>
Okay, the user wants me to explain why keyword research is important for SEO. They need a short paragraph followed by three practical benefits. Let me start by recalling what I know about SEO. Keyword research is the foundation of any SEO strategy. It helps in understanding what users are searching for, which is crucial for content creation and optimization.

First, I should define keyword research briefly. It involves identifying the keywords that people use when searching for information, products, or services. This helps in aligning content with user intent. Then, the importance part: it ensures that the content is relevant and meets the needs of the target audience. This improves the chances of the website being found by search engines and users.

Now, the three practical benefits. The first one is improved content relevance. By targeting the right keywords, the content becomes more aligned with what users are looking for. Second, better SEO performance. Using the right keywords can lead to higher rankings in search results. Third, increased organic traffic. When content is optimized for relevant keywords, it attracts more traffic from users who are interested in the topic.

Wait, but I need to make sure these points are concise. Let me check if there's a better way to phrase them. Maybe for the first benefit, focus on content relevance and user intent. Second, optimizing for keywords helps with on-page SEO elements like title tags and meta descriptions. Third, it helps in identifying long-tail keywords which can capture more specific searches and lead to higher conversion rates.

Wait, but the user asked for three practical benefits. Let me make sure each is a distinct benefit. Maybe the first is content relevance, the second is SEO performance, and the third is traffic. Alternatively, maybe the third is about targeting specific audiences. I need to be clear and concise. Also, the user might be a content creator or a marketing professional looking to understand the value of keyword research. They might not have a deep technical background, so the explanation should be straightforward. Avoid jargon but still be precise. Make sure the paragraph is one short paragraph, not too long, and the three points are separate. Let me structure that.
</think>

Keyword research is essential for SEO because it helps identify the specific terms and phrases users search for, enabling content and website optimization to align with user intent and demand. By understanding these keywords, businesses can create targeted content, improve search engine rankings, and attract relevant traffic.

Practical Benefits:
1. **Improved Content Relevance:** Ensures content addresses actual user needs and questions, increasing the likelihood of engagement and conversion.
2. **Enhanced SEO Performance:** Targets high-value keywords that improve visibility in search results, boosting organic rankings.
3. **Targeted Audience Acquisition:** Focuses on specific keywords that attract users with clear interests, leading to more effective marketing efforts.

==========
Temperature: 1.1
Top-K: 40
Top-P: 0.90
Time: 76.07 seconds

Response:
<think>
Okay, the user is asking for an explanation of why keyword research is important for SEO. They want a short paragraph followed by three practical benefits. Let me start by recalling what I know about keyword research. It's a fundamental part of SEO, right? It helps in understanding what users are searching for, which is crucial for content creation and optimization.

I need to explain the importance in one paragraph. Maybe start with how keyword research is the foundation of SEO strategies. It helps align content with user intent and search trends. That's important because it ensures that the website is optimized for the right keywords, which can improve visibility and rankings.

Now, the three practical benefits. First, better content creation. If you know the keywords users are searching for, you can create content that addresses their needs. Second, targeted audience. Using relevant keywords helps attract the right audience, which can lead to higher engagement and conversions. Third, improved ROI. By focusing on the right keywords, you can optimize your efforts and get better returns from your SEO efforts.

Wait, let me make sure those benefits are accurate. Yes, each of those points is a common benefit of keyword research. Maybe check if there's a more specific example. For instance, instead of "targeted audience," maybe "improved user experience" or "better targeting." But the user asked for three practical benefits, so the three I have are solid. Need to make sure they're concise and directly tied to SEO success.

Also, the user might be a marketing or content professional looking to understand the value of keyword research. They might be trying to justify SEO efforts or decide on strategies. So, the answer should be clear and actionable. The paragraph should encapsulate the main idea, and the benefits should highlight the tangible outcomes of doing keyword research.

I think that's all. Let me structure the answer as a short paragraph followed by the three points. Make sure the paragraph is concise and the benefits are practical and directly related to SEO outcomes.
</think>

Keyword research is essential for SEO because it helps identify the terms and phrases users search for, aligning content with their intent and optimizing visibility in search results. By understanding target keywords, businesses can create content that addresses specific needs, improves rankings, and drives organic traffic.

****Three practical benefits:****
1. **Targeted audience engagement – Focused keywords attract users actively seeking solutions, increasing the likelihood of conversions.**
2. **Improved content relevance – Ensures content is tailored to searchers' queries, enhancing user experience and satisfaction.**
3. **Efficient resource allocation – Prioritizes efforts on high-impact keywords, saving time and budget while maximizing ROI.**

==========
Temperature experiment completed.

Temperature-Only Test – Output Analysis

In this experiment, only **Temperature** was changed while **Top-K remained 40** and **Top-P remained 0.90**. This helps us observe the effect of Temperature independently.

Temperature	Response Time	Observed Output	Variation
0.2	74.04 s	More conventional and specific. Discussed content relevance, specific audiences, and long-tail keywords.	Low
0.7	80.05 s	Balanced and clear. Discussed content relevance, SEO performance, and targeted audiences.	Medium
1.1	76.07 s	More varied ideas. Introduced efficient resource allocation, saving budget, and maximizing ROI.	Higher

8.2 Top-K-Only Test

```
top_k_values = [10, 40, 100]

for value in top_k_values:
    output, response_time = generate_response(
        TEST_PROMPT,
        temperature=0.7,
        top_k=value,
        top_p=0.90
```

```
    )

    print("\nTop-K:", value)
    print("Time:", response_time, "seconds")
    print(output)
```

Expected effect: a small Top-K restricts token choices, while a larger Top-K may increase wording variation.

Qwen Output :

```
======================================================================
CONTROLLED EXPERIMENT: TOP-K
Fixed Temperature = 0.7
Fixed Top-P = 0.90
======================================================================

======================================================================
Temperature: 0.7
Top-K: 10
Top-P: 0.90
Time: 58.99 seconds

Response:
<think>
Okay, the user is asking to explain why keyword research is important for SEO, with one short paragraph and three practical benefits. Let me start by recalling the basics of SEO. Keyword research is a foundational part of SEO because it helps in understanding what users are searching for. It's not just about finding keywords, but aligning content with user intent.

First, I need to make sure the paragraph covers the main points: how keyword research helps in targeting the right audience, improving content relevance, and optimizing for search engines. Then, the three benefits should be practical. Let me think of the key benefits.

The first benefit could be improved content relevance. By knowing the keywords, you can create content that matches user needs. Second, maybe enhanced visibility. Using the right keywords can help your site rank higher in search results. Third, better user experience. Aligning content with search intent ensures the site is useful and relevant, leading to a better user experience.

Wait, are those all benefits? Let me check. Improved content relevance, enhanced visibility, and better user experience. That seems solid. Need to make sure each benefit is concise and directly tied to keyword research. Also, the user might be a beginner in SEO, so the explanation should be clear without jargon. Avoid being too technical. Make sure the paragraph is a single, concise statement. Alright, that should cover it.
</think>

Keyword research is essential for SEO because it helps identify the specific terms and phrases users search for, allowing businesses to align their content with search intent and optimize for visibility. By understanding what audiences are looking for, websites can create targeted, relevant content that meets user needs and improves search engine rankings.

**Practical benefits:**
1. **Improved content relevance** – Ensures content is tailored to user queries, increasing engagement and conversion.
2. **Enhanced visibility** – Targets high-traffic keywords to boost search engine rankings and organic traffic.
3. **Better user experience** – Aligns content with user intent, making the site more intuitive and valuable for visitors.

======================================================================
Temperature: 0.7
Top-K: 40
Top-P: 0.90
Time: 57.63 seconds

Response:
<think>
Okay, the user is asking me to explain why keyword research is important for SEO, with one short paragraph and three practical benefits. Let me start by recalling what I know about SEO and keyword research.

First, keyword research is crucial because it helps identify what users are searching for, which aligns content with user intent. That's the main point. Then, the benefits... Well, the first benefit is that it helps in creating targeted content. By understanding the keywords, you can tailor your content to meet specific needs, which improves relevance and, consequently, rankings.

Next, the second benefit is improving the user experience. If you know the keywords your audience is using, you can make sure your website is optimized for those terms, which makes navigation easier and more intuitive. That leads to better engagement and lower bounce rates.

Third, keyword research can help in identifying gaps in the competition. By analyzing which keywords competitors are targeting, you can find opportunities to target less competitive keywords, which might be more effective for your site. This allows for a more strategic approach to SEO.
```

Wait, but the user wants three practical benefits. Let me check if I have three. Yes, targeted content, user experience, and competitive advantage. That should cover it. I need to make sure the paragraph is concise and the benefits are clearly listed. Also, the user might be an SEO professional or someone starting in the field, so the explanation should be clear and to the point without jargon. Make sure the paragraph is one sentence, and the benefits are three bullet points. Alright, that should do it.
</think>

Keyword research is essential for SEO because it helps identify the terms and phrases users search for, enabling content to align with their intent and improve visibility in search results. It ensures that websites focus on relevant topics, optimize for user needs, and stay competitive in a dynamic digital landscape.

Practical Benefits:
1. **Targeted Content Creation:** Identifies specific keywords to tailor content, improving relevance and engagement.
2. **Improved User Experience:** Aligns website structure and navigation with user search behavior, reducing bounce rates.
3. **Competitive Advantage:** Reveals gaps in competitors' strategies, allowing for niche or less competitive keyword targeting.

==========
Temperature: 0.7
Top-K: 100
Top-P: 0.90
Time: 63.68 seconds

Response:
<think>
Okay, the user wants me to explain why keyword research is important for SEO. They need a short paragraph followed by three practical benefits. Let me start by recalling the basics of SEO. Keyword research is a foundational part of SEO because it helps identify what users are searching for. It allows websites to align their content with user intent, which is crucial for visibility.

Now, the main point is to emphasize that keyword research is essential for understanding user needs and optimizing content. The first paragraph should be concise, highlighting how keyword research helps in targeting the right keywords, aligning content with user intent, and improving search engine rankings.

Next, the three practical benefits. The first one could be improved visibility through targeted content. Second, it helps in creating content that meets user needs, leading to higher engagement. Third, it allows for efficient use of resources by focusing on high-value keywords. Wait, is there another benefit? Maybe optimizing for conversions? Or maybe improving content relevance? Let me think. The three benefits should be distinct. Maybe: 1. Targeting the right keywords for better visibility. 2. Enhancing user experience by addressing their needs. 3. Efficient resource allocation by focusing on high-impact keywords. Yeah, that makes sense. Need to make sure each benefit is clear and practical. Let me check if these points are accurate. Yes, keyword research helps in targeting the right keywords, which leads to better rankings. It also ensures content is relevant to users, improving engagement. And it helps in using resources wisely by focusing on keywords that have high potential. That's three solid points. I need to make sure the paragraph is a single concise statement and the benefits are each a separate point. Alright, that should cover it.
</think>

Keyword research is crucial for SEO because it helps identify the terms and phrases users search for, aligning a website's content with search intent and improving visibility in search engine results. By understanding what audiences are looking for, businesses can optimize their content to meet specific needs, enhance relevance, and attract organic traffic.

Practical benefits:
1. **Improved visibility:** Targeting high-value keywords increases the chances of appearing in search results, driving more traffic to the site.
2. **Enhanced user experience:** Content tailored to search intent meets user needs, leading to higher engagement and lower bounce rates.
3. **Efficient resource allocation:** Focusing on keywords with high potential and low competition ensures resources are spent on strategies that deliver measurable results.

==========
Top-K experiment completed.

Top-K-Only Test – Output Analysis

In this experiment, only **Top-K** was changed while **Temperature remained 0.7** and **Top-P remained 0.90**. This allows us to observe the effect of Top-K independently.

Top-K	Response Time	Observed Output	Variation
10	58.99 s	More straightforward and conventional. Focused on content relevance, visibility, and user experience.	Low

Top-K	Response Time	Observed Output	Variation
40	57.63 s	More varied. Introduced targeted content creation and competitive advantage.	Medium
100	63.68 s	Broader ideas. Introduced efficient resource allocation and measurable results.	Higher

8.3 Top-P-Only Test

```
top_p_values = [0.50, 0.90, 0.99]

for value in top_p_values:
    output, response_time = generate_response(
        TEST_PROMPT,
        temperature=0.7,
        top_k=40,
        top_p=value
    )

    print("\nTop-P:", value)
    print("Time:", response_time, "seconds")
    print(output)
```

Expected effect: lower Top-P should be more conservative, while higher Top-P should allow more diverse output.

Qwen Output :

==========
CONTROLLED EXPERIMENT: TOP-K
Fixed Temperature = 0.7
Fixed Top-P = 0.90
==========

==========
Temperature: 0.7
Top-K: 10
Top-P: 0.90
Time: 58.99 seconds

Response:
<think>
Okay, the user is asking to explain why keyword research is important for SEO, with one short paragraph and three practical benefits. Let me start by recalling the basics of SEO. Keyword research is a foundational part of SEO because it helps in understanding what users are searching for. It's not just about finding keywords, but aligning content with user intent.

First, I need to make sure the paragraph covers the main points: how keyword research helps in targeting the right audience, improving content relevance, and optimizing for search engines. Then, the three benefits should be practical. Let me think of the key benefits.

The first benefit could be improved content relevance. By knowing the keywords, you can create content that matches user needs. Second, maybe enhanced visibility. Using the right keywords can help your site rank higher in search results. Third, better user experience. Aligning content with search intent ensures the site is useful and relevant, leading to a better user experience.

Wait, are those all benefits? Let me check. Improved content relevance, enhanced visibility, and better user experience. That seems solid. Need to make sure each benefit is concise and directly tied to keyword research. Also, the user might be a beginner in SEO, so the explanation should be clear without jargon. Avoid being too technical. Make sure the paragraph is a single, concise statement. Alright, that should cover it.
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Next, the second benefit is improving the user experience. If you know the keywords your audience is using, you can make sure your website is optimized for those terms, which makes navigation easier and more intuitive. That leads to better engagement and lower bounce rates.

Third, keyword research can help in identifying gaps in the competition. By analyzing which keywords competitors are targeting, you can find opportunities to target less competitive keywords, which might be more effective for your site. This allows for a more strategic approach to SEO.

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Practical Benefits:
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1. **Improved visibility:** Targeting high-value keywords increases the chances of appearing in search results, driving more traffic to the site.

2. **Enhanced user experience:** Content tailored to search intent meets user needs, leading to higher engagement and lower bounce rates.
3. **Efficient resource allocation:** Focusing on keywords with high potential and low competition ensures resources are spent on strategies that deliver measurable results.

==========
Top-K experiment completed.
(venv) PS D:\SpecialTopicsinAI> python top_p_test.py
==========
CONTROLLED EXPERIMENT: TOP-P
Fixed Temperature = 0.7
Fixed Top-K = 40
==========

==========
Temperature: 0.7
Top-K: 40
Top-P: 0.5
Time: 49.31 seconds

Response:
<think>
Okay, the user wants me to explain why keyword research is important for SEO. They need a short paragraph followed by three practical benefits. Let me start by recalling what keyword research entails. It's about identifying the words and phrases that people use when searching for something online. For SEO, this is crucial because it helps in creating content that aligns with what users are looking for.

So, the main point is that keyword research helps in understanding user intent and needs. That's why it's important for SEO. Now, the paragraph should summarize that. Then, the three benefits. Let me think. First, it helps in creating targeted content that matches user search intent. Second, it improves the relevance and quality of the content, which is a key factor in search engine rankings. Third, it allows for better optimization of on-page elements like titles and meta descriptions, which can enhance click-through rates and overall SEO performance.

Wait, are those the most practical benefits? Maybe. Let me check. Yes, those three points cover the main aspects: content relevance, quality, and on-page optimization. I should make sure each benefit is concise and directly tied to SEO. Also, the paragraph should be a single sentence that encapsulates the importance. Let me put it all together now.
</think>

Keyword research is essential for SEO because it helps identify the specific terms and phrases users search for, enabling content creators to align their strategies with user intent and improve visibility in search results.
Practical benefits:
1. **Targeted content creation:** It ensures content is optimized for relevant keywords, increasing the likelihood of higher rankings and engagement.
2. **Improved relevance:** Understanding search terms helps tailor content to meet user needs, enhancing both user experience and search engine rankings.
3. **Efficient resource allocation:** It prioritizes high-impact keywords, saving time and effort by focusing on terms with the most potential to drive traffic and conversions.

==========
Temperature: 0.7
Top-K: 40
Top-P: 0.9
Time: 75.93 seconds

Response:
<think>
Okay, the user is asking why keyword research is important for SEO. I need to explain that in one short paragraph and then list three practical benefits. Let me start by recalling what keyword research entails. It's about identifying the words and phrases people use when searching for something online. For SEO, this is crucial because it helps in creating content that matches user intent, which in turn improves search engine rankings.

So the main point is that keyword research helps align content with what users are actually searching for. That makes the content more relevant, leading to better visibility. Now, the paragraph should be concise. Maybe start with the importance of understanding user intent and how that aligns with SEO strategies.

Now, the three benefits. First, it helps in creating content that targets the right keywords, which can increase organic traffic. Second, it identifies high-traffic, low-competition keywords that can be leveraged for better rankings. Third, it allows for optimizing on-page elements like titles and meta descriptions, which are crucial for search engines to understand the page's content.

Wait, are there other benefits? Maybe improving user experience by aligning content with user needs. But the user asked for three practical ones. Let me make sure each benefit is clear and actionable. Also, need to avoid jargon so the explanation is accessible. Let me structure the paragraph first, then the three points. Check if the benefits are distinct and practical. Yes, the three points I listed seem solid. Make sure each is a benefit that's directly tied to SEO performance.
</think>

Keyword research is essential for SEO because it helps identify the specific words and phrases users search for, allowing content creators to align their strategies with user intent and improve visibility in search results. By understanding what terms people use, businesses can optimize their content, websites, and marketing efforts to target relevant audiences more effectively.

****Practical Benefits:****
1. **Improved Content Relevance: Ensures content is tailored to user needs, increasing the likelihood of higher engagement and conversion rates.**
2. **Targeted Traffic: Helps attract high-intent visitors by focusing on keywords that align with specific goals or problems.**
3. **Competitive Advantage: Reveals gaps in competitors' strategies, enabling optimized content that outperforms rival sites in search rankings.**

==========
Temperature: 0.7
Top-K: 40
Top-P: 0.99
Time: 60.49 seconds

Response:
<think>
Okay, the user wants an explanation of why keyword research is important for SEO. They need a short paragraph followed by three practical benefits. Let me start by recalling the main points about keyword research. It's about understanding what users are searching for, which helps in creating content that meets their needs. This aligns with SEO goals because it ensures the content is relevant and targeted.

Now, the first paragraph should summarize the importance. Maybe start with how keyword research helps in targeting the right audience, improving content relevance, and optimizing for search engines. Then mention that it's a foundational step for SEO strategies.

Next, the three benefits. The first benefit could be improved content relevance and user experience. By using keywords that users actually search for, the content is more likely to address their needs, leading to better engagement and lower bounce rates.

Second benefit: enhanced search engine rankings. SEO algorithms prioritize content that matches user intent, so using effective keywords helps the site rank higher in search results, increasing visibility.

Third benefit: better resource allocation. With keyword research, you can focus on high-traffic, high-intent keywords, making your efforts more efficient and cost-effective. This helps in directing marketing resources where they have the most impact.

Wait, is there a fourth benefit? Maybe not necessary. Need to ensure the three are distinct and practical. Let me check if each benefit is clearly tied to SEO outcomes. Yes, relevance, rankings, and resource efficiency. That should cover it. Now, structure the paragraph and the benefits clearly without being too technical. Keep it concise and to the point.
</think>

****Why keyword research is important for SEO:****
Keyword research is the foundation of any effective SEO strategy, as it helps identify the specific terms and phrases users search for, enabling content to align with their intent and needs. By understanding these keywords, businesses can create targeted content, optimize on-page elements, and improve visibility in search results, ultimately driving organic traffic and conversions.

****Practical benefits:****
1. **Improved content relevance: Ensures content is tailored to user intent, enhancing user experience and reducing bounce rates.**
2. **Enhanced search engine rankings: Targets keywords that search engines prioritize, increasing the likelihood of higher rankings for relevant queries.**
3. **Better resource allocation: Focuses efforts on high-traffic, high-intent keywords, maximizing the efficiency of SEO campaigns.**

==========
Top-P experiment completed.

Top-P-Only Test – Output Analysis

In this experiment, only **Top-P** was changed while **Temperature remained 0.7** and **Top-K remained 40**. This allows the individual effect of Top-P to be observed more clearly.

Top-P	Response Time	Observed Output	Variation
0.50	49.31 s	More focused and conventional. Discussed targeted content, relevance, and resource allocation.	Low
0.90	75.93 s	More varied. Introduced targeted traffic and competitive advantage.	Medium

Top-P	Response Time	Observed Output	Variation
0.99	60.49 s	Broader and slightly more detailed. Discussed user experience, rankings, resource allocation, organic traffic, and conversions.	Higher

9. Complete sampling_benchmark.py Program

```
import os
import time
import requests
import pandas as pd
from dotenv import load_dotenv

load_dotenv()

QWEN_URL = os.getenv("QWEN_URL")
MODEL_NAME = os.getenv("MODEL_NAME")

TEST_PROMPT = (
    "Explain why keyword research is important for SEO. "
    "Give one short paragraph followed by three practical benefits."
)

def generate_response(prompt, temperature, top_k, top_p):
    payload = {
        "model": MODEL_NAME,
        "prompt": prompt,
        "stream": False,
        "options": {
            "temperature": temperature,
            "top_k": top_k,
            "top_p": top_p
        }
    }

    start_time = time.time()

    response = requests.post(
        QWEN_URL,
        json=payload,
        timeout=300
    )

    end_time = time.time()

    response.raise_for_status()
    result = response.json()

    return result["response"], round(end_time - start_time, 2)

experiments = [
    {"setting": "Conservative", "temperature": 0.2, "top_k": 20, "top_p": 0.70},
    {"setting": "Balanced", "temperature": 0.7, "top_k": 40, "top_p": 0.90},
    {"setting": "Creative", "temperature": 1.1, "top_k": 80, "top_p": 0.98}
```

```
]

results = []

for experiment in experiments:
    output, response_time = generate_response(
        TEST_PROMPT,
        experiment["temperature"],
        experiment["top_k"],
        experiment["top_p"]
    )

    results.append({
        "Setting": experiment["setting"],
        "Temperature": experiment["temperature"],
        "Top-K": experiment["top_k"],
        "Top-P": experiment["top_p"],
        "Time (seconds)": response_time,
        "Response": output
    })

    print("\n" + "=" * 70)
    print("Setting:", experiment["setting"])
    print("Temperature:", experiment["temperature"])
    print("Top-K:", experiment["top_k"])
    print("Top-P:", experiment["top_p"])
    print("Time:", response_time, "seconds")
    print("\nResponse:")
    print(output)

df = pd.DataFrame(results)

print("\nExperiment Results:")
print(df[[
    "Setting",
    "Temperature",
    "Top-K",
    "Top-P",
    "Time (seconds)"
]])
```

QWEN OUTPUT:

```
======================================================================
```

Setting: Conservative
Temperature: 0.2
Top-K: 20
Top-P: 0.7
Time: 103.03 seconds

Response:
<think>
Okay, the user is asking me to explain why keyword research is important for SEO. They want a short paragraph followed by three practical benefits. Let me start by recalling what I know about SEO and keyword research.

First, keyword research is crucial because it helps identify what users are searching for, which aligns content with user intent. That's the main point. Then, I need to think of three practical benefits. Let me brainstorm. Maybe improving content relevance, optimizing for better rankings, and increasing organic traffic. Wait, but I need to make sure these are specific and practical.

Wait, the first benefit could be about aligning content with user intent, which leads to better engagement. Second, it helps in optimizing on-page elements like titles and meta tags, which can improve search engine rankings. Third, it allows for targeted content creation, which can drive more organic traffic.

But I need to check if these are accurate. Also, the user might be a content creator or a marketer looking to improve their SEO strategy. They might not be familiar with the technical aspects, so the explanation should be clear and concise.

I should make sure the paragraph is a single, coherent statement that covers the importance. Then list the three benefits in a way that's easy to understand. Let me structure the paragraph first: keyword research helps identify the terms users are searching for, which allows content to be tailored to their needs, improving relevance and engagement. Then the three benefits: 1. Enhances content relevance and user experience. 2. Improves search engine rankings through optimized on-page elements. 3. Drives targeted organic traffic to the website.

Wait, is that three? Yes. Let me check if each benefit is a practical one. Yes, they all relate to tangible outcomes. I think that's solid. Make sure the paragraph is one sentence, but maybe a bit more detailed. Wait, the user said "one short paragraph," so maybe a couple of sentences. Let me rephrase the paragraph to be more concise.

"Keyword research is essential for SEO because it helps identify the specific terms and phrases users are searching for, allowing content to be tailored to their needs and intent, which improves relevance, engagement, and search engine rankings." Then the three benefits.

Wait, but the user might want the three benefits to be separate points. Let me make sure each is a clear benefit. Maybe:

1. Enhances content relevance and user experience by aligning with search intent.
2. Improves search engine rankings through optimized on-page elements like titles and meta tags.
3. Drives targeted organic traffic to the website, increasing visibility and conversion opportunities.

Yes, that seems good. I think that's the answer.
</think>

Keyword research is essential for SEO because it helps identify the specific terms and phrases users are searching for, allowing content to be tailored to their needs and intent, which improves relevance, engagement, and search engine rankings.

****Practical Benefits:****
1. **Enhances content relevance and user experience by aligning with search intent, ensuring content meets actual user needs.**
2. **Improves search engine rankings through optimized on-page elements like titles, meta tags, and headers, which signal to search engines what the page is about.**
3. **Drives targeted organic traffic to the website, increasing visibility and conversion opportunities by focusing on high-intent keywords.**

==========
Setting: Balanced
Temperature: 0.7
Top-K: 40
Top-P: 0.9
Time: 82.08 seconds

Response:
<think>
Okay, the user is asking me to explain why keyword research is important for SEO. They want a short paragraph followed by three practical benefits. Let me start by recalling what I know about keyword research. SEO is all about improving a website's visibility in search engines, right? So keyword research is the process of identifying the words and phrases that people use when searching for something. That makes sense because if you know what people are searching for, you can optimize your content to match those terms.

Now, why is that important? Well, if you target the right keywords, your content is more likely to be found by the intended audience. That's a key point. But I need to make sure I explain that in a concise paragraph. Then, the three practical benefits. Let me think. The first benefit is probably improving content relevance. By understanding the keywords, you can create content that actually answers the user's needs, which helps with ranking. Second, maybe increasing targeted traffic. If you use the right keywords, your site will be more visible in search results, leading to more traffic. Third, optimizing for long-term SEO. Using the right keywords helps in building a strong foundation for your SEO strategy, which can lead to sustained results.

Wait, but are there other benefits? Maybe improving user experience? Or maybe increasing conversion rates? Hmm, but the user asked for three practical benefits. I should stick to the three I have. Let me check if those are accurate. Yes, relevance, targeted traffic, and long-term optimization. That seems solid. I need to make sure the paragraph is concise and the benefits are clear and practical. Alright, that should cover it.
</think>

Keyword research is crucial for SEO because it helps identify the specific terms and phrases users search for, allowing websites to align their content with user intent and improve visibility in search results. By understanding what audiences are looking for, businesses can optimize their content, structure, and strategies to attract relevant traffic and enhance rankings.

****Practical Benefits:****
1. **Improves Content Relevance: Targets keywords that align with user needs, ensuring content addresses the right queries and increases the likelihood of higher rankings.**
2. **Increases Targeted Traffic: Attracts high-intent visitors interested in specific topics, leading to better engagement and conversion opportunities.**
3. **Optimizes Long-Term SEO Strategy: Builds a foundation for sustainable SEO growth by focusing on keywords that drive consistent, quality traffic over time.**

==========
Setting: Creative
Temperature: 1.1
Top-K: 80
Top-P: 0.98
Time: 60.52 seconds

Response:
<think>
Okay, the user is asking me to explain why keyword research is important for SEO. They want a short paragraph and three practical benefits. Let me start by recalling what keyword research entails. It's about identifying the words and phrases that people use when searching for something online. For SEO, this is crucial because it helps in aligning content with what users are actually looking for.

First, I need to make sure the paragraph covers the main points. SEO is about improving visibility on search engines, and keyword research is the foundation. It helps in creating content that matches user intent, which is key for rankings. Then, I need to break it down into three practical benefits. Let me think: one benefit could be optimizing content for the right audience, leading to better engagement. Another might be improving website rankings by aligning with search terms. A third could be enhancing user experience by addressing specific queries, which in turn increases conversions or traffic.

Wait, does that cover all the key points? Maybe I should check if there's any overlap or if I need to make it more concise. Also, the user might be a business owner or a marketing professional looking to understand the value of keyword research. They probably want to know how it directly impacts their SEO efforts and their business outcomes. So, the benefits need to be actionable and show tangible results. I should ensure each benefit is distinct and clearly linked to SEO success. Let me structure the paragraph first, then list the benefits, making sure they are specific and practical.
</think>

Keyword research is essential for SEO because it helps identify the specific words and phrases users search for, allowing websites to align their content with user intent and improve visibility in search results. By understanding these terms, businesses can optimize their content, metadata, and strategies to target relevant audiences, ultimately driving more organic traffic and conversions.

****Practical Benefits:****
1. **Targeted Content Creation: Identifies high-potential keywords to create content that resonates with specific audience needs, increasing engagement and relevance.**
2. **Improved Rankings: Enhances website performance by aligning with search terms that search engines prioritize, boosting visibility in SERPs.**
3. **Cost Efficiency: Focuses resources on keywords with measurable impact, avoiding costly experiments and ensuring efforts are directed toward proven opportunities.**

Experiment Results:

	Setting	Temperature	Top-K	Top-P	Time (seconds)
0	Conservative	0.2	20	0.70	103.03
1	Balanced	0.7	40	0.90	82.08
2	Creative	1.1	80	0.98	60.52

10. What to Observe

- Determinism: Does the model give similar wording when a low-randomness setting is used?
- Creativity: Does the model provide more varied words, examples or sentence structures at higher values?
- Relevance: Does the response stay focused on the question?
- Structure: Does it continue to follow the requested paragraph plus three benefits format?
- Consistency: Are the answers stable or do they change significantly between runs?
- Response time: Is there any noticeable difference in generation time?

11. Comparative Results Table

Fill this table after running the test prompt. The observations should be based on the actual outputs produced on your computer.

Setting	Temperature	Top-K	Top-P	Response Time (s)	Determinism	Creativity	Structure	Hallucination	Main Observation
Conservative	0.2	20	0.70	**103.03**	High	Low	Highly structured and direct	No obvious hallucination observed	Produced the most controlled and predictable response. It closely followed the requested format and used conventional SEO benefits.
Balanced	0.7	40	0.90	**82.08**	Medium	Medium	Clear and well structured	No obvious hallucination observed	Maintained good structure while introducing slightly broader ideas, such as long-term SEO strategy. It provided a balance between consistency and variation.
Creative	1.1	80	0.98	**60.52**	Lower	High	Structured, but more varied in	No obvious hallucination	Produced the most varied response and

Setting	Temperature	Top-K	Top-P	Response Time (s)	Determinism	Creativity	Structure	Hallucination	Main Observation
							content	observed	introduced a less obvious benefit, “Cost Efficiency.”

12. Discussion

The experiment showed noticeable differences in the responses generated using the three sampling configurations. Although all three responses remained relevant, factually reasonable, and followed the requested structure, their level of variation changed as the Temperature, Top-K, and Top-P values increased.

The **Conservative setting** used a Temperature of **0.2**, Top-K of **20**, and Top-P of **0.70**. It produced the most controlled and conventional response. The answer focused on common SEO benefits such as content relevance, improved rankings, and targeted organic traffic. This behavior was consistent with the expected effect of lower sampling values, where the model selects from a narrower range of highly probable tokens.

The **Balanced setting**, with Temperature **0.7**, Top-K **40**, and Top-P **0.90**, maintained a clear and structured response while showing somewhat greater variation. For example, it introduced **long-term SEO strategy** as one of the practical benefits. This suggests that moderate sampling values provided a balance between predictable generation and diversity.

The **Creative setting** used the highest values: Temperature **1.1**, Top-K **80**, and Top-P **0.98**. It showed the greatest content variation and introduced **cost efficiency** as a benefit, which was not included in the other two responses. This matches the expected behavior because higher Temperature increases randomness, higher Top-K provides more candidate tokens, and higher Top-P allows the model to consider a broader probability distribution.

No obvious hallucinations were observed in any of the three responses. All remained relevant to keyword research and SEO. The recorded response times were **103.03 seconds** for Conservative, **82.08 seconds** for Balanced, and **60.52 seconds** for Creative. However, these results are not sufficient to conclude that higher sampling values improve generation speed because execution time can also be affected by hardware load, model warm-up, response length, and other runtime conditions.

Overall, the observed results generally matched the expected behavior of the sampling parameters. However, higher values were not necessarily better. Controlled settings are more suitable for factual and technical tasks where consistency is important, whereas higher sampling values can be useful for brainstorming and creative tasks where greater variation is desirable.

13. Conclusion

This project demonstrated how **Temperature, Top-K, and Top-P** affect the generation behavior of the locally running **Qwen model through Ollama**. Temperature controls the randomness of token selection, Top-K restricts selection to a specified number of high-probability tokens, and Top-P restricts token choices according to their cumulative probability.

The experimental results showed that the **Conservative configuration** generated a more controlled and conventional response, the **Balanced configuration** provided a compromise between consistency and variation, and the **Creative configuration** generated more diverse content while still maintaining the required structure and relevance.

No obvious hallucinations were identified in the tested responses. The results therefore demonstrate that sampling parameters can significantly influence how an LLM expresses an answer even when the prompt remains unchanged. The most appropriate configuration depends on the intended task: lower and moderate sampling values are generally preferable when consistency is important, while higher values can be useful when creativity and variation are desired.

For a stronger evaluation of **determinism**, the same configuration should be executed multiple times and the resulting responses compared.